LAB 5

#1

```
#include <iostream>
using namespace std;

class Base
{
public:
    void showBase()
    {
        cout << "Base class function\n";
    }
};

class Derived : public Base
{
public:
    void showDerived()
    {
        cout << "Derived class function\n";
    }
};

int main()
{
    Base b;
    Derived d;

    b.showBase();
    d.showBase();
    d.showDerived();

    return 0;
}
```

```
Base class function
Base class function
Derived class function
```

#2

```
#include <iostream>
using namespace std;

class Base
{
protected:
    int x;
public:
    Base()
    {
        x = 10;
    }
};
```

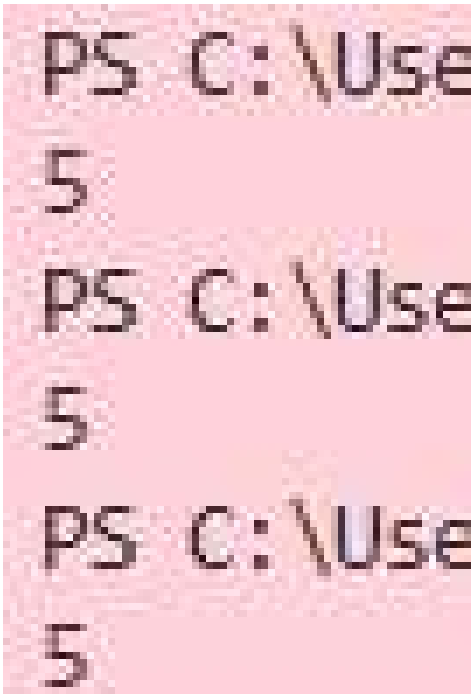
```
Value of x: 10
```

```
class Derived : public Base
{
public:
    void display()
    {
        cout << "Value of x: " << x << endl;
    }
};

int main()
{
    Derived d;
    d.display();
    return 0;
}
```

#3

//PUBLIC:

```
#include <iostream>
using namespace std;

class A
{
public:
    int x = 5;
};

class B : public A
{
};

int main()
{
    B obj;
    cout << obj.x << endl;
    return 0;
}
```



//PROTECTED:

```
#include <iostream>
using namespace std;

class A
{
public:
    int x = 5;
};

class B : protected A
{
public:
    void show()
```

```
    {
        cout << x << endl;
    }
};

int main()
{
    B obj;
    obj.show();
    return 0;
}
```

//PRIVATE:

```
#include <iostream>
using namespace std;

class A
{
public:
    int x = 5;
};

class B : private A
{
public:
    void show()
    {
        cout << x << endl;
    }
};

int main()
{
    B obj;
    obj.show();
    return 0;
}
```

#4

//SINGLE

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B : public A
{
};
```

```
PS C:\Users\lenovo\De
Class A
```

```
int main()
{
    B obj;
    obj.showA();
    return 0;
}
```

```
PS C:\Users\lenovo\De
Class A
Class B
```

//MULTIPLE:

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B
{
public:
    void showB()
    {
        cout << "Class B\n";
    }
};
class C : public A, public B
{
};

int main()
{
    C obj;
    obj.showA();
    obj.showB();
    return 0;
}
```

```
Class A
```

//MULTILEVEL:

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B : public A
{
};
class C : public B
{
```

```
};

int main()
{
    C obj;
    obj.showA();
    return 0;
}
```

//Hierarchical:

```
#include <iostream>
using namespace std;
class A
{
public:
    void show()
    {
        cout << "Base Class\n";
    }
};
class B : public A
{
};
class C : public A
{
};
int main()
{
    B b;
    C c;
    b.show();
    c.show();

    return 0;
}
```

```
PS C:\Users\le
Base Class
Base Class
```

//HYBRID:

```
#include <iostream>
using namespace std;
class A
{
public:
    void showA()
    {
        cout << "Class A\n";
    }
};
class B : public A
{
};
class C : public A
{
};
class D : public B, public C
```

```
Class A
Class A
```

```
{
};
int main()
{
    D obj;
    obj.B::showA();
    obj.C::showA();
    return 0;
}
```

#5

```
#include <iostream>
using namespace std;
class Base
{
public:
    Base()
    {
        cout << "Base Constructor\n";
    }
    ~Base()
    {
        cout << "Base Destructor\n";
    }
};
class Derived : public Base
{
public:
    Derived()
    {
        cout << "Derived Constructor\n";
    }
    ~Derived()
    {
        cout << "Derived Destructor\n";
    }
};

int main()
{
    Derived obj;
    return 0;
}
```

```
Base Constructor
Derived Constructor
Derived Destructor
Base Destructor
```

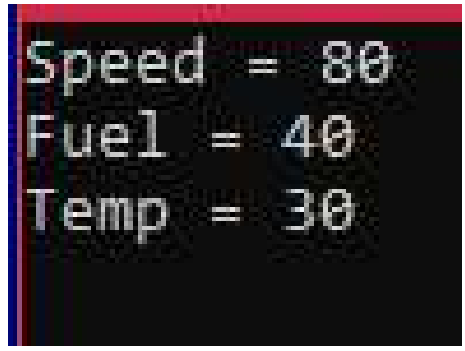
#6

```
#include <iostream>
using namespace std;
class Book
{
protected:
    string title;
    string author;
    float price;
public:
    Book(string t, string a, float p)
    {
        title = t;
        author = a;
        price = p;
    }
    void displayBook()
    {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Price: " << price << endl;
    }
};

class Textbook : public Book
{
    string subject;

public:
    Textbook(string t, string a, float p, string s) :
Book(t, a, p)
    {
        subject = s;
    }

    void displayTextbook()
    {
        displayBook();
        cout << "Subject: " << subject << endl;
    }
};

int main()
{
    Textbook t("C++ Programming", "Bjarne",
500, "Programming");
    t.displayTextbook();
    return 0;
}
```

```
Title: C++ Programming
Author: Bjarne
Price: 500
Subject: Programming
```

#7

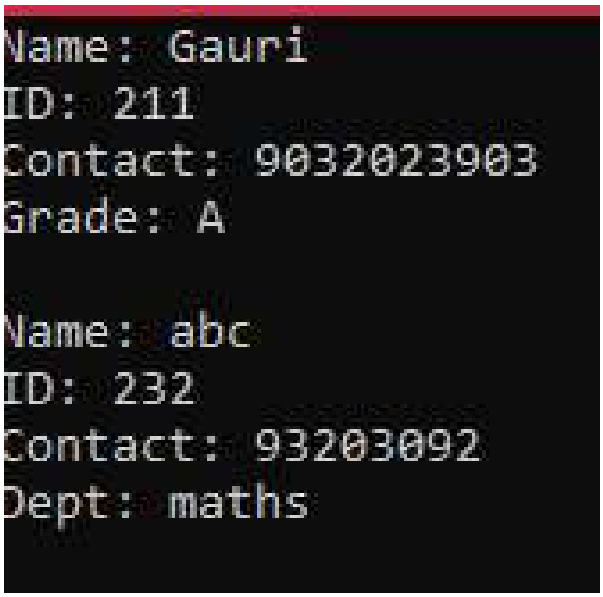
```
#include <iostream>
using namespace std;
class Speedometer
{
	protected:
		int speed;
	public:
		Speedometer(int s)
		{
			speed = s;
		}
		void showspeed()
		{
			cout<<"Speed =
"<<speed<<endl;
		}
};
class fuelgauge
{
	protected:
		int fuel;
	public:
		fuelgauge(int f)
		{
			fuel = f;
		}
		void showfuel()
		{
			cout<<"Fuel =
"<<fuel<<endl;
		}
};
class thermo
{
	protected:
		int temp;
	public:
		thermo(int t)
		{
			temp=t;
		}
		void showtemp()
		{
			cout<<"Temp =
"<<temp<<endl;
		}
};
class dash:  public Speedometer,public
fuelgauge, public thermo
{
	public:
		dash(int s,int f, int t) :
Speedometer(s),fuelgauge(f),thermo(t)
		{
```



```
		}
		void dispdash()
		{
			showspeed();
			showfuel();
			showtemp();
		}
};
int main()
{
	dash car(80,40,30);
	car.dispdash();
	return 0;
}
```

#8

```
#include <iostream>
using namespace std;
class libuser
{
	protected:
		string name;
		int id;
		string contact;
	public:
		libuser(string n, int i, string c)
		{
			name=n;
			id=i;
			contact=c;
		}
		void disp()
		{
			cout<<"Name: "<<name<<endl;
			cout<<"ID: "<<id<<endl;
			cout<<"Contact: "<<contact<<endl;
		}
};
class student:public libuser
{
	string grade;
public:
	student(string n, int i, string c, string g): libuser(n,i,c)
	{
		grade = g;
	}
	void dispstud()
	{
		disp();
		cout<<"Grade: "<<grade<<endl;
	}
```



```
};
class teacher: public libuser
{
	string dept;
public:
	teacher(string n, int i, string c, string d) :
libuser(n,i,c)
	{
		dept=d;
	}
	void dispteach()
	{
		disp();
		cout<<"Dept: "<<dept<<endl;
	}
};
int main()
{
	student
s("Gauri",211,"9032023903","A");
	teacher
t("abc",232,"93203092","maths");
	s.dispstud();
	cout<<endl;
	t.dispteach();
	return 0;
}
```

#9

```
#include <iostream>
using namespace std;
class vehicle
{
	protected:
		string make;
		int year;
		string model;
	public:
		vehicle(string m, int y, string mo)
		{
			make=m;
			year=y;
			model=mo;
		}
		void disp()
		{
			cout<<"Make:
"<<make<<endl;
			cout<<"Model:
"<<model<<endl;
			cout<<"Year:
"<<year<<endl;
		}
};
```

```
Make: Tata
Model: xyz
Year: 2000
Load capactiy = 50000000
Temp control: -18

--------------------------------
```

```
class truck:public vehicle
{
protected:
	int loadcap;
public:
	truck(string m,int y,string mo,int lc):
vehicle(m,y,mo)
	{
		loadcap=lc;
	}
	void dispt()
	{
		disp();
		cout<<"Load capactiy =
"<<loadcap<<endl;
	}
};
class rt: public truck
{
	int tempc;
public:
	rt(string m,int y,string mo,int lc,int tc) :
truck(m,y,mo,lc)
	{
		tempc=tc;
	}
	void disprt()
	{
		dispt();
		cout<<"Temp control:
"<<tempc<<endl;
	}
};
int main()
{
	rt t1("Tata",2000,"xyz",50000000,-18);
	t1.disprt();
	return 0;
}
```

#10

```
#include <iostream>
using namespace std;
class person
{
	protected:
		string name;
		string address;
	public:
		person(string n,string a)
		{
			name=n;
			address=a;
		}
		void disp()
		{
			cout<<"Name: 
"<<name<<endl;
			cout<<"Address: 
"<<address<<endl;
		}
};
class staff:virtual public person
{
protected:
	int empid;
	string dep;
public:
	staff(string n,string a,int id,string d):
person(n,a)
	{
		empid=id;
		dep=d;
	}
	void dispstaf()
	{
		disp();
		cout<<"Employee id: 
"<<empid<<endl;
		cout<<"Department: 
"<<dep<<endl;
	}
};
class student: virtual public person
{
protected:
	int studid;
	string grade;
public:
	student(string n,string a,int id,string 
g):person(n,a)
	{
		studid=id;
		grade=g;
	}
	void dispstud()
```

```
Name: ABC
Address: city
Name: ABC
Address: city
Employee id: 3902
Department: CSED
student id: 3920
grade: A
```

```
	{
		cout<<"student id:
"<<studid<<endl;
		cout<<"grade: "<<grade<<endl;
	}
};
class ta: public staff,public student
{
public:
	ta(string n,string a,int empid,string
dep,int sid,string
g):person(n,a),staff(n,a,empid,dep),student(n,a,
sid,g)
	{
	}
	void dispta()
	{
		disp();
		dispstaf();
		dispstud();
	}
};
int main()
{
	ta
ta1("ABC","city",3902,"CSED",3920,"A");
	ta1.dispta();
	return 0;
}
```